

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CE/IT)

Sem - IV University Examination, 2010-11

CE – 210 Microprocessor & Computer Architecture

Date: 14.05.11, Saturday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 70

- Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

Section-I

Q.1(a) Attempt all question

05

- (1) Which signal is used to demultiplex lower address bus from the time multiplexed data bus in 8085?
- (2) When 8085 is RESET, it starts fetching its first instruction from _____ address.
- (3) Number of machine cycles in an instruction depends on the size of the instruction in 8085. State true/false and justify.
- (4) If memory mapped I/Os are used, IO/M* will never go high in 8085 system during the data transfer between processor and I/Os. State true/false and justify.
- (5) 13 address lines are needed to identify 16K registers in a memory chip. True/false and justify. (Show your calculation also)

Q.1(b) Describe the meaning of following 8085 pins.

06

ALE, IO/M, X1 and X2, RST7.5, Hold, HLDA

Q.2(a) Write a program to add only positive readings from the ten 8-bit readings which are stored at memory address 2300h to 2309H. Store the result at memory location 230Ah. Draw neat flow chart and explain your logic clearly. State your assumptions clearly, if any.

06

Q.2(b) 2100 LXI H, 2300H

06

LXI SP, 3000H

P.T.O.

LXI B,1234H

PUSH B

XTHL

POP B

What will be the content of B, C, H, L & SP registers after the execution of POP B instruction. Show the stack frame.

OR

Q.2(a) Write an assembly language program to count number of '1's in an 8 bit number which is at memory location 2200H. Store the result at 2201H. Draw neat flow chart and explain your logic very clearly. 06

Q.2(b) Attempt all question 06

(1) State the differences between opcode fetch and memory read cycle.

(2) When you are comparing two 8-bit numbers using CMP instructions, how you will treat carry flag to interpret the result? Explain.

(3) If 8085 is operating at 1 MHz clock, how much time 8085 will take to transfer one byte information to peripheral device connected as I/O mapped I/O? Show your calculation.

Q.3(a) Draw and explain the timing diagram of IN 00H. 06

Q3 (b) Explain the working of CALL and RET instructions. 06

OR

Q.3(a) LXI B,FFFFH 06

LOOP: DCXB

JNZ LOOP

NOP

Find the mistake in the above program and modify the program so that when BC reaches to zero, program should come out from the loop.

Q.3(b) (1) If 2 Kbytes memory chip is mapped to last 2 Kbytes address space of 8085, What is the starting and last address? 06

(2) If the device connected on the system bus is slow, explain the

support provided by 8085 to synchronize its speed with the device speed.

(3) State the function of the following instructions and content of the register(s) after the execution of the instruction :

(i) LXI H,2300h (ii) LDAX D

Section-II

Q.4(a) Distinguish Hardware and Software interrupt. Explain interrupt vector diagram. 06

Q.4(b) Explain transmitter and receiver section in UART 8251 IC. 05

Q.5(a) Give comparison of Master and Slave mode in 8257 IC. 06

Q.5(b) What is key-debounce? Give the software and hardware techniques to solve the key-debounce problem. 06

OR

Q.5(a) Describe Mode 1 of 8255 with help of examples and diagrams. 06

Q.5(b) Discuss the initialization of 8259 IC. 06

Q.6(a) Explain logic block diagram of 80386. 06

Q.6(b) Write a short note on Pentium processor 06

OR

Q.6(a) Give the features of Pentium IV processor and explain with diagram. 05

Q.6(b) (1) Draw and explain 8086 microprocessor with its flag register. 05

(2) Give the difference between two major processor Pentium III and Pentium IV 02